Efficient and Unbiased Safety Test for Autonomous Driving Systems

Zhengmin Jiang, Wenbo Pan, Jia Liu, Shaobo Dang, Zhiheng Yang, Huiyun Li, *Senior Member, IEEE*, and Yi Pan, *Senior Member, IEEE*

***Abstract*—Test the safety of Autonomous Driving Systems (ADS) with realistic traffic conditions is important to the insurance industry, legislators, and third-party technical services. However, the scarcity of risky driving events distributed in real-world driving often makes sampling inefficient. In this paper, we propose a unified and hierarchical testing framework for efficient and unbiased safety tests of ADS. We first extract the risk subspace from the Naturalistic Driving Data (NDD) with defined safety measures. Subsequently, we use the Gaussian Copula method to accurately model the subspace joint probability density function (PDF). We further formulate a Kriging model-based optimization problem for finding the appropriate Importance Sampling (IS) parameters, where the Kriging model is trained iteratively within limited computational resources. Eventually, the risky concrete scenarios library is generated using the obtained Importance Sampling parameters. With the simulation results skewered back to the original risk subspace, the ADS probability of failure (e.g., crash rate) can thus be accurately and unbiasedly estimated. Experiment results show that using 0.94% computational resources, the Kriging surrogate model captures 96.04% of ADS crashes with at least 90.25% precision. On crash rate estimation, our proposed method achieves a consistent result with that of Monte Carlo simulation, provided that the relative error does not exceed 10% while improving the testing efficiency by up to 6,539 times.**

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Zhengmin Jiang is with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: jiangzm@siat.ac.cn).

Wenbo Pan is with the CRRC Zhuzhou Institute Co, Ltd., Zhuzhou 412001, China (e-mail: panwb1@csrzic.com).

Jia Liu, Shaobo Dang, and Zhiheng Yang are with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China (e-mail: jia.liu1@siat.ac.cn; shaobo.dang@siat.ac.cn; zh.yang@siat.ac.cn).

Huiyun Li is with the Shenzhen Institute of Advanced Technology, and Guangdong-Hong Kong-Macao Joint Laboratory of Human-Machine Intelligence-Synergy Systems, Chinese Academy of Sciences, Shenzhen 518055, China (e-mail: hy.li@siat.ac.cn).

Yi Pan is with the Dean of Computer Science and Control Engineering, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China (e-mail: yi.pan@siat.ac.cn).

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***Index Terms*—Autonomous driving system, naturalistic driving data, risk subspace, unbiased test, efficiency.**

I. Introduction

AUTONOMOUS Driving System (ADS) is promising to solve the problems of traffic congestion and fuel emission. However, recent traffic accidents involving autonomous vehicles severely eroded public confidence in this technology [1], [2]. Consequently, thorough tests on ADS safety before market launch become compelling demand.

Approaches for ADS testing can be divided into two main categories: physical test and virtual test as shown in Fig. 1. The physical test mainly includes closed-field test and open road test, providing a comprehensive but time-consuming measure for autonomous vehicles operating in the real world [3]. The virtual test mimics the real world and vehicles with digital models and preset simulation scenarios, which have the advantages of high efficiency and low cost. Li Li et al. presented a new semantic diagram to design driving tasks for ADS evaluation [4], [5] and proposed a parallel testing system [6]. This system integrates physical test and virtual test tightly to validate the perception, planning, and control capabilities of autonomous vehicles. In general, with flexible configuration and high efficiency, virtual testing has become the mainstream approach for the safety test of ADS.

The virtual test can be further divided into knowledge-based, machine learning-based, and safety-centric based approaches. The knowledge-based approach relies on traffic regulations or domain knowledge to create a library of simulation scenarios. The main methods in this category include ontology, test matrix, combinatorial testing, etc. Bagschik et al. and Elgharbawy et al. used ontology to generate structured terms for scenarios, then transform the terms into the parameter space, and finally obtained simulation scenarios [7], [8]. The test matrix refers to a table of predefined parameter permutations for test procedures, such as ASSESS [9]. The parameters are often taken from Naturalistic Driving Data (NDD), accidents database, or test standards. However, the size of the test matrix expands exponentially when adding parameters, as a result, the test cost will increase. Combinatorial Test can greatly reduce the number of test cases in the test matrix while ensuring coverage [10], [11]. The authors in [12] proposed a scenario complexity-based combinatorial test method to improve the efficiency in finding failures of Lane Departure Warning systems. Generally speaking, the test cases generated by the knowledge-based approach

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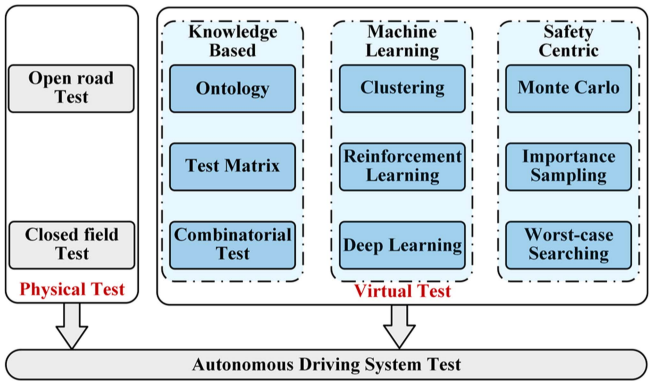


Fig. 1. Summary of test approaches for ADS.

are fixed and predefined. Therefore the latent failure modes of ADS may not be effectively revealed.

Machine learning approaches, such as unsupervised learning, reinforcement learning, and deep learning, have also been used to generate simulation scenarios. S. Li et al. applied auto-encoder and k-means to categorize real-world driving encounters into distinguishable simulation scenario clusters [13]. Langner et al. also employed an auto-encoder to recognize novel scenarios within the driving data sequence [14]. These investigators can only filter test cases of interest from existing data. Alternatively, reinforcement learning can be utilized to develop scenarios through the interaction between ADS and its environment. For example, Kochenderfer et al. introduced the Adaptive Stress Testing (AST) framework to detect the failures of ADS, as described in [15], [16], [17]. In addition, several other scholars have sought to study the utilization of deep learning models. Jenkins et al. automatically generated new accident scenarios based on V2X data using RNN and showed that the generated scenarios were similar to real accidents [18]. Demetriou et al. suggested a Recurrent Conditional-Generative Adversarial Network to derive driving trajectories of varying lengths [19].

Assessing ADS safety with realistic traffic conditions is important to the insurance industry, legislators, and third-party technical services. To this end, safety-centric sampling from naturalistic traffic conditions is often used to determine the collision risk [20], such as Monte Carlo sampling [21], [22] and Importance Sampling (IS). Zhao [23], [24] employed Importance Sampling to skew the original probability density and increase the exposure rate of rare events. Huang et al. further proposed piecewise mixture models instead of single parametric distribution models [25]. Xu et al. used genetic algorithms to calculate the optimal IS parameters [26]. S. Feng et al. divided the scenario parameters into critical and non-critical and used Importance Sampling for critical parameters and Monte Carlo sampling for non-critical parameters, thus balancing sampling efficiency and estimation accuracy [27], [28], [29]. In addition, worst-case searching algorithms have been studied by Raja et al. [30], Tuncali et al. [31], and Klischat et al. [32] to identify the riskiest scenarios. Nevertheless, it is inefficient in searching for ADS defects in high-dimensional parameter space.

In summary, the virtual test has attracted far more interest, where how to proactively obtain challenging scenarios is the key to testing the autonomous vehicle [33]. However, until now, existing methods remain some issues:

1) *Low efficiency:* The knowledge-based method relies on predefined rules to abstract scenarios, while Monte Carlo sampling usually suffers from low efficiency due to the huge test space. Therefore, even if significant testing resources have been allocated, the performance of ADS under challenging conditions is poorly accounted for.
2) *Incompatibility with realism:* Machine learning and worst-case search often provide collections of risky scenarios that can cause ADS to fail. However, a common criticism is that the possibility of occurrence in the real world of the generated test scenarios is not considered [3].
3) *Biased test results:* The Importance Sampling technique can derive risky but reasonable scenarios from NDD. However, NDD is often an enormous-sized dataset with high-dimensional and coupled variables. For simplicity, the decision variables identified by NDD are usually treated as independent, and this assumption may bias the test results. Moreover, how to obtain the IS proposal distribution cost-effectively has not yet been fully discussed.

To address these issues, a unified and hierarchical framework for ADS safety test is proposed in this paper, as shown in Fig. 2. The four steps are integrated to explore the ADS probability of failure under challenging and realistic traffic conditions in an efficient and unbiased manner: (1) Risk subspace is set up using domain knowledge, driving events cleaning and clustering. (2) The joint probability density function (PDF) of the decision variables determined by the risk subspace is modeled. (3) A Kriging model-based optimization problem is formulated for finding the appropriate IS distribution parameters, where the Kriging model is trained iteratively within limited computational resources. (4) Risk test cases (or named concrete scenarios in PEGASUS [34]) that may lead to autonomous vehicle crashes are further generated. With the generated risk test cases, ADS is simulated and evaluated.

The remainder of this article is organized as follows: Section II.A introduces the details of how to extract risk subspace from NDD, and Section II.B further models the joint PDF of decision variables using the Gaussian Copula approach. Section III.A introduces the theoretical basis of the probability of failure estimation. Section III.B presents the Kriging model and searches for reasonable IS distribution parameters efficiently. Using the obtained Importance Sampling distribution, risky concrete scenarios are generated, simulated, and analyzed in Section IV. Some related research issues are discussed in Section V. Finally, we conclude this paper in Section VI.

II. Naturalistic Driving Data Processing

The process for extracting the risk subspace from the naturalistic driving dataset is shown in Fig. 3.

A. Identification of the Risk Subspace

In this paper, we use the highD naturalistic driving data [35] for risk subspace extraction and ADS test. Due to the limited detection range of ADS, relative distance larger than 150 meters,

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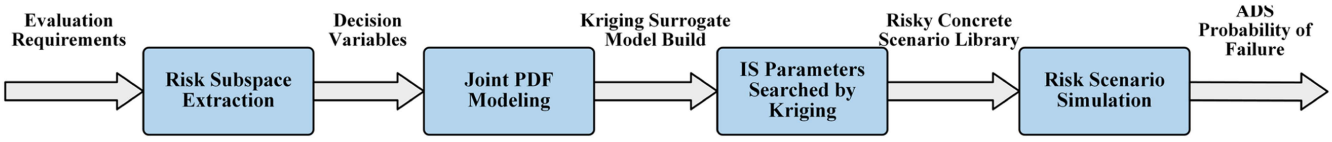


Fig. 2. A unified and hierarchical framework for efficient and unbiased safety test for ADS. PDF is short for Probability Density Function. IS stands for Importance Sampling.

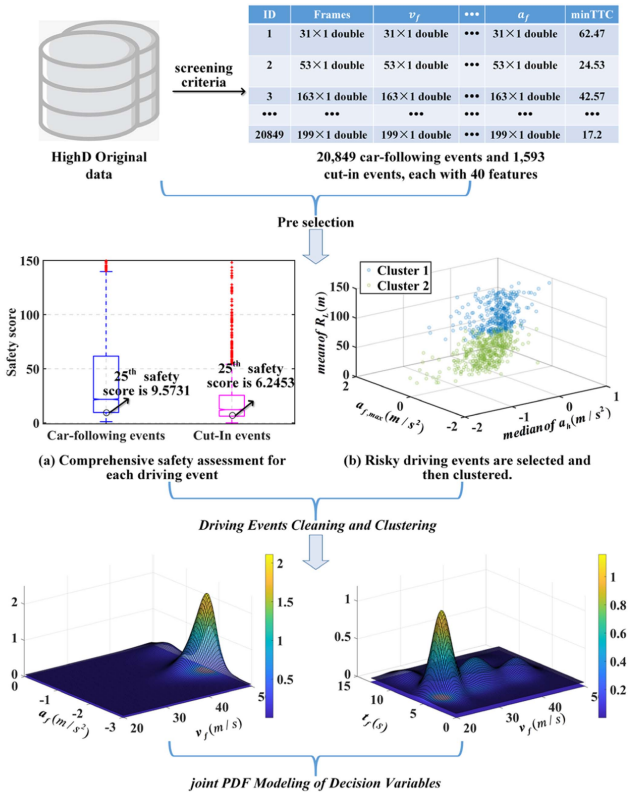


Fig. 3. Cleaning, clustering, and risk subspace setup process of naturalistic driving data. v_f (m/s), a_f (m/s^2) And t_f (s) Are the velocity, deceleration, and deceleration duration of the leading vehicle. a_h (m/s^2) is the acceleration of the ego vehicle. R_L (m) is the spacing between the ego car and the adversary vehicle.

time to collision (TTC) less than zero, or trucks' data were filtered out, we summarize the data screening criteria as follows:

- Driving data of trucks is filtered out
- $0.1\text{s} \leq \text{TTC} \leq 10\text{s}$
- If a lane change occurs, the data fragment between the lane change starting point and ending point is taken as a cut-in event.
- Otherwise, a data fragment where the relative following distance is 0.1m∼150m is considered a car-following event.
- Only the driving events with more than 25 consecutive recorded values are valid (the highD sampling frequency is 25 Hz).

1) Driving Events Cleaning: Investigating the most challenging driving events is the crucial step in setting up the risk subspace. Currently, a variety of surrogate safety measures (SSM) are used to uncover risky driving events [36], [37]. However, it is easy to underestimate the driving risk when only a single type of SSM is used.

In this paper, besides the commonly used safety measures, such as Time-based SSM (TTC and THW), we extend the safety factors to include the distance safety factor (ξ) and the inverse DRAC (DRAC^{-1}, Deceleration to avoid a crash):

$$\begin{cases} \xi = (R_L - R_{brake})/(R_{safe} - R_{brake}) \\ DRAC^{-1} = (2 \times R_L)/(v_h - v_f)^2 \end{cases} \tag{1}$$

where R_L (m) is the relative spacing between the ego car and the adversary vehicle. R_{brake} (m) and R_{safe} (m) are the braking distance and safety distance, respectively. v_h (m/s) and v_f (m/s) are the speed of the ego car and the leading vehicle, respectively. Using the entropy weight theory [38], we synthesize the above safety factors into a comprehensive safety assessment. The risky driving events are thus screened if the overall safety assessment is less than a set threshold (here the threshold can be adjusted according to the needs of research).

2) Feature Selection and Clustering: Notice that the risk subspace can be further mined by clustering, however, the above filtered risky driving events are high-dimensional time series data comprising 40 features. A forward likelihood ratio test of logistic regression is thus used to extract driving event features, followed by k-means clustering. The logistic regression model of whether a driving event is dangerous or not is

$$\ln\left(\frac{p}{1-p}\right) = \alpha + \sum_{i=1}^{n} \beta_i m_i \tag{2}$$

where p is the occurrence ratio of risky driving events. m is the variable that has a significant effect on the overall safety assessment, α is the regression intercept, and β is the partial regression coefficient.

B. Joint Probability Density Function Modeling

The parameters determined by the risk subspace are taken as a vector of decision variables as $\boldsymbol{x} = [x_1, x_2, \ldots, x_d]$, where d denotes the dimensionality of $\boldsymbol{x}$. Taking the car-following scenario as an example, the decision variables can be represented as

$$\boldsymbol{x} = [v_f, a_f, t_f, R_L] \tag{3}$$

where a_f (m/s^2) and t_f (s) are the deceleration and deceleration time of the leading vehicle.

To estimate the ADS probability of failure by simulation, the main task is to derive concrete scenarios using the decision variables. To this end, the joint PDF of decision variables is

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usually demanded. Most existing studies tend to assume these decision variables are independent of each other, therefore test results are often biased. Some other research tends to use Gaussian Mixture Models (GMM) to construct a multivariate PDF [39], [40], [41]. Essentially, GMM requires iteration with the Expectation Maximization algorithm, which suffers from a long training time when dealing with large and high-dimensional data. In this study, the Gaussian Copula method is applied to model the joint PDF without the assumption of independent variables.

The Gaussian Copula decomposes the joint PDF into the marginal distributions of each random variable and a Copula function. The Copula function is used to describe the interactions between the decision variables. This method only needs to deal with marginal distributions of individual variables and does not require an iterative process, so it is not sensitive to the data size.

According to Sklar's theorem [42], the joint distribution $F(\boldsymbol{x})$ of any *d* random variables can be decomposed into the marginal distributions $F(x_1), \ldots, F(x_d)$ and a Copula function.

$$\begin{aligned} F(\boldsymbol{x}) &= C\left(F\left(x_1\right), F\left(x_2\right), \ldots, F\left(x_d\right)\right) \\ &= C\left(u_1, u_2, \ldots, u_d\right) \end{aligned} \tag{4}$$

where $C(\cdot)$ is a copula function. $F(x_i) = u_i$ is the cumulative density function of the ith decision variable x_i. The joint PDF $f(\boldsymbol{x})$ is shown as

$$f(\boldsymbol{x}) = c\left(u_1, u_2, \ldots, u_d\right) \cdot f\left(x_1\right) \ldots f\left(x_d\right) \tag{5}$$

where $c(\cdot)$ is the PDF of the Copula function. $f(x_i)$ is the marginal probability density distribution, which can be fitted by parametric or nonparametric distribution models. For the Gaussian Copula used in this study [43].

$$\begin{aligned} c\left(u_1, u_2, \ldots, u_d\right) = & \frac{\partial^d C\left(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_d)\right)}{\partial \Phi^{-1}(u_1) \cdots \partial \Phi^{-1}(u_d)} \\ & \times \frac{1}{\phi\left(\Phi^{-1}(u_1)\right)} \cdots \frac{1}{\phi\left(\Phi^{-1}(u_d)\right)} \end{aligned} \tag{6}$$

where $\phi(\cdot)$ and $\Phi^{-1}(\cdot)$ are the probability density function and inverse cumulative density function of a standard normal distribution.

In short, through driving events cleaning and feature selection, it is feasible to set up the risk subspace through clustering, wherein each point in the cluster corresponds to a time-series driving event. Moreover, the steps for modeling the joint PDF of the decision variables are summarized as follows.

1) Estimating the marginal probability distribution for each decision variable.
2) Using the cumulative function of the fitted marginal distribution, convert the data to the copula scale (unit square).
3) Fitting a Gaussian Copula to the data.

III. Efficient and Unbiased Evaluation

In this section, we first introduce the theoretical foundations of unbiased estimation of the ADS probability of failure, such as crash rate. To accelerate the search for reasonable Importance Sampling distribution parameters, we then present Kriging models to interpolate the test results. Finally, the small size but challenging candidate concrete scenario library is generated using the searched Importance Sampling distribution. In other words, with the help of the Kriging model and Importance Sampling theory, the number of actual executions of the time-consuming and high-fidelity simulation model is greatly reduced. Moreover, the ADS crash rate in the candidate library can be skewed back to the risk subspace while the mathematical expectation is consistent with that of Monte Carlo simulation (i.e., the unbiasedness is guaranteed).

A. Crash Rate Estimation

We formulate the indicator function of whether a concrete scenario results in an ADS crash as:

$$I\left(\boldsymbol{x}_i\right) = \begin{cases} 1, & \boldsymbol{x}_i \; will \; lead \; to \; acollision \; test \; result \\ 0, & otherwise \end{cases} \tag{7}$$

From an information theory point of view, the uncertainty of concrete scenarios can be described by the joint entropy.

$$H(\boldsymbol{x}) = -\sum_{x_i \in \chi_i} f\left(x_1, x_2, \ldots, x_d\right) \cdot \log_2 f(x_1, x_2, \ldots, x_d) \tag{8}$$

where χ_i is the feasible set of x_i and $H(x_i)$ is the information entropy. $f(x_i)$ is the marginal probability distribution of x_i. $H(\boldsymbol{x})$ has the following relation with $H(x_i)$

$$H(\boldsymbol{x}) \leq H(x_1) + H(x_2) + \ldots + H(x_d) \tag{9}$$

According to the maximum entropy theory [44], $H(\boldsymbol{x})$ reaches its maximum value when the decision variable x_i obeys a uniform distribution on χ_i. That is, when we randomly sample the decision variables uniformly and combine them randomly, the test cases will theoretically cover all possible outcomes.

In practice, $f(x_i)$ is not uniformly distributed. $H(\boldsymbol{x})$ in such a case is less than the theoretical maximum entropy, but a sufficient number of ADS crashes can still be observed. This way of generating concrete scenarios according to the joint PDF $f(\boldsymbol{x})$ is also known as the Monte Carlo sampling.

According to the Strong Law of Large Numbers, the Monte Carlo estimation ($\hat{\gamma}$) obey the following relationship with the ground truth (γ).

$$P\left(\lim_{N \to \infty} \hat{\gamma} = \gamma\right) = 1 \tag{10}$$

Where N is the number of test cases. In practical applications, N is usually large enough, we can then calculate the ADS probability of failure using the average of the indicator function happening, approximately [23].

$$\gamma \approx \frac{1}{N} \sum_{i=1}^{N} I\left(\boldsymbol{x}_i\right), \quad \boldsymbol{x}_i \sim f\left(\boldsymbol{x}\right) \tag{11}$$

Correspondingly, derived from the Central Limit Theorem, the estimation error of γ is described in the following

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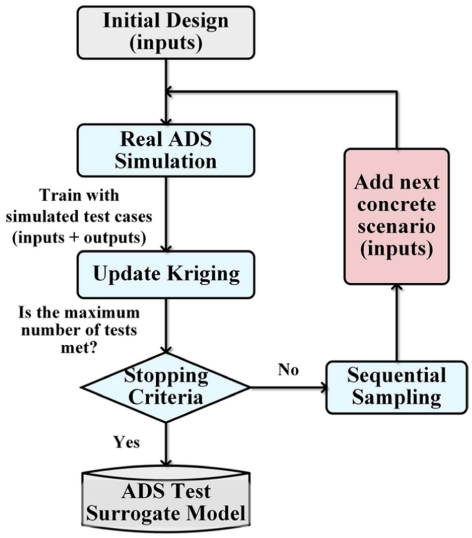


Fig. 4. The general framework of training an ADS test surrogate model.

equation [45].

$$e_{MC} = \frac{1}{N}\sqrt{\left(\frac{1}{\gamma}-1\right)^2 N\cdot\gamma + N\cdot(1-\gamma)} = \sqrt{\frac{1-\gamma}{N\cdot\gamma}} \tag{12}$$

Generally speaking, Monte Carlo can theoretically help us to calculate the ADS crash rate in a brute-force solution, Nevertheless, the lower the probability of failure, the larger N , and such a large test scale is easy to make the computational overload.

Importance Sampling (IS), as an efficient variance reduction technique, can significantly reduce the number of simulations while maintaining the estimation accuracy. The core idea of Importance Sampling is to replace the original density function $f\left(\boldsymbol{x}\right)$ with a new density $h\left(\boldsymbol{x}\right)$ that has a higher likelihood of the rare event happening. The crash rate can be rewritten as

$$\gamma \approx \frac{1}{N}\sum_{i=1}^{N} I\left(\boldsymbol{x}_i\right) L\left(\boldsymbol{x}_i\right), \quad \boldsymbol{x}_i \sim h(\boldsymbol{x}) \tag{13}$$

where $L(\boldsymbol{x}) = f\left(\boldsymbol{x}\right)/h\left(\boldsymbol{x}\right)$ is the likelihood ratio. The IS estimation error [46] is

$$e_{IS} = \frac{1}{N}\sqrt{\sum_{i=1}^{N}\left(\frac{I\left(\boldsymbol{x}_i\right) f\left(\boldsymbol{x}_i\right)}{E\left(I(\boldsymbol{x})\right) h\left(\boldsymbol{x}_i\right)} - 1\right)^2}, \quad \boldsymbol{x}_i \sim h(\boldsymbol{x}) \tag{14}$$

where $E(I(\boldsymbol{x}))$ is the mathematical expectation of $I\left(\boldsymbol{x}\right)$. To achieve the optimal $h\left(\boldsymbol{x}\right)$ as accurately as possible, a large number of high-fidelity simulation runs are usually required.

B. Search for Reasonable IS Parameters

1) Kriging Surrogate Model: The Kriging surrogate model makes it feasible to capture the Best Linear Unbiased Prediction of the unknown test case within limited resources. As stated in Fig. 4, the ADS Kriging surrogate model is not built in a one-shot manner.

In this paper, the minimum relative distance between the autonomous vehicle and the traffic car is taken as the output in the Kriging model. The smaller the distance, the higher the danger and a collision occur when the distance is negative.

A sequential sampling strategy that maximizes the acquisition in eq. (15) is used to iteratively select training test cases in the feasible sets.

$$acquision = \begin{cases} (\hat{y}_{\max} - \hat{y})\cdot Dis_{\min} + s\cdot\phi\left(\frac{\hat{y}_{\min}-\hat{y}}{s}\right) & s > 0 \\ 0 & s = 0 \end{cases} \tag{15}$$

where $\hat{y}$ and s are the predicted value and variance of the Kriging model, respectively. ϕ is the probability density function of the standard normal distribution. Dis_{min} is the spatial distance between the candidate test case and the trained sets. Note that the training samples in the acquisition function must be normalized.

It can be concluded that those candidate test cases that are far away from the existing trained sets or have a higher probability of being dangerous, or a larger estimation error, will lead to an increase in acquisition value. A more detailed mathematical proof of the Kriging surrogate model can be referenced in [47].

2) Search for Reasonable IS Parameters: The parameters of the IS distribution $h\left(\boldsymbol{x}\right)$ are often searched iteratively, where costly batch simulations are performed in each iteration. The lower the frequency of a rare event, the slower the search with the greater computational burden.

We transform the problem of searching for reasonable IS parameters into an optimization problem. That is, the Kriging model is used to reduce the time-consuming simulations to determine the parameters at a low cost. Take the multivariate Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \sum)$ as the candidate IS distribution. Consider the following optimization problem.

$$\left(\boldsymbol{\mu}^*, \sum^*\right) = \underset{(\boldsymbol{\mu},\sum)\in G\times H}{argmin}\ \frac{1}{N}\sqrt{\sum_{i=1}^{N}\left(\frac{I\left(\boldsymbol{x}_i\right) f\left(\boldsymbol{x}_i\right)}{E\left(I(\boldsymbol{x})\right) h\left(\boldsymbol{x}_i\right)} - 1\right)^2} \tag{16}$$

where $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$ are the mean vector and covariance matrix of the kth searched IS distribution parameters, respectively. In (16), the objective is to lower the crash rate estimation error. *G* and *H* are search domains. The general workflow is summarized as Algorithm 1.

In short, finding the reasonable IS parameters does not invoke iterated simulations, so the search can be done at a very high speed. Moreover, the Kriging model can be used repeatedly for any arbitrary changes in optimization objectives or constraints.

IV. Experimental Results

Car-following and cut-in scenarios are chosen as case studies in this paper. The corresponding risk subspaces in NDD are first mined, and the joint PDFs are further modeled. Kriging surrogate models are then established within limited computational resources. Comparative experiments are conducted to validate the Kriging surrogate model can indeed accelerate the ADS test while maintaining accuracy. Afterward, the reasonable Importance Sampling distributions parameters are determined,

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Algorithm 1: Kriging based IS distribution Searching

Inputs: Feasible sets of decision variables, Initial training set is sampled by the Latin Hypercube Sampling strategy [48]

Output: Reasonable IS distribution parameters: $\mathcal{N}(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*)$

```
Constructing the initial Kriging model M_0;
While (The number of iterations < maximum iteration times threshold α_1) do
    x_new ← Select the next training case that maximizes equation (15);
    Simulate the response value at x_new and merge it into the training set;
    Update Kriging model;
Return the trained surrogate model M*
While (e_IS > estimation error threshold α_2) do
    μ_k, Σ_k ← Optimize equation (16) in search domains;
    x ~ N(μ_k, Σ_k) Generate a certain number of test cases based on the current kth IS distribution;
    Predict the response value for each x based on M*;
    γ ← Calculate the estimated ADS crash rate using equation (13);
    e_IS ← Update the estimated relative error using (14);
return μ*, Σ*
```

and the ADS crash rates within different risk subspaces are investigated.

A. Risk Subspace Setup

According to the data screening criteria described in Section II, 20849 valid car-following events and 1593 valid cut-in events were selected from the highD dataset, respectively.

The entropy weight method [38] is used to determine the weights of the various safety factors with the following steps.

- Standardize the safety factors.
- Determine the variance of the standardized safety factors.
- Calculate the information entropy of each indicator.
- If the information entropy of a safety factor is low, assign it a higher weight.

Finally, the comprehensive safety scores are calculated as follows.

$$\begin{cases} Score_1 = w_1 \cdot \left[TTC \quad THW \quad \xi \quad DRAC^{-1}\right]^T \\ Score_2 = w_2 \cdot \left[TTC \quad THW \quad \xi \quad DRAC^{-1}\right]^T \\ w_1 = [0.2147 \quad 0.0510 \quad 0.0401 \quad 0.6942] \\ w_2 = [0.2077 \quad 0.0159 \quad 0.0501 \quad 0.7263] \end{cases} \tag{17}$$

where TTC (s) and THW (s) represent Time To Collision and Time Headway, respectively. ξ and DRAC^{-1} ($m^{-1}s^2$) are the distance safety factor and the inverse Deceleration to Avoid a Crash, respectively. w_1, w_2 are the weight vectors for calculating the comprehensive safety scores of car-following ($Score_1$) and cut-in ($Score_2$).

We thus labeled a driving event as dangerous if the comprehensive safety score is below the 25th percentile. Furthermore, the logistic regression fitting information is shown in Table I.

TABLE I
LOGISTIC REGRESSION ANALYSIS RESULTS

Model	-2 log-likelihood	Sig.	R^2	Degree of freedom
Car-following	190.78	0.0001	0.996	10
Cut-in	440.05	0.0002	0.847	5

Table I illustrates that the logistic regression models can fit the labeled naturalistic driving events well (the significance level of both models was less than 0.05 while the Nagelkerke R^2 reached 0.996 and 0.847, respectively). Using the forward likelihood ratio test analysis, ten and five statistical indicators from the original 40-dimensional raw data are subsequently selected for car-following and cut-in clustering, respectively.

The silhouette score is used in this paper to determine the best number of clusters. Generally speaking, the more its value converges to one, the better the clustering result. The silhouette score in Fig. 5(a) reaches its maximum when the number of clusters is taken as two, thus the risky car-following and cut-in driving events can be separated into two clusters respectively. Fig. 5(b) shows that the car-following cluster-1 summarizes mainly the risky driving events of large relative distance and velocity; whereas car-following cluster-2 contains more events of the lead vehicle slowing down. Similarly, it can also be concluded that the cut-in cluster-1 describes the traffic vehicle performing lane change with a smaller initial spacing.

Further, variables such as speed (v_f , *m/s*), deceleration (a_f, *m/s*^{2}) and deceleration time (t_f, *s*) are chosen as the parameters describing the motion of the leading vehicle, while the relative distance (R_L, *m*) is also used as the decision variable to derive concrete scenarios. Fig. 6 shows the part joint PDFs of the four decision variables using the Gaussian Copula method (see Section II.B).

It is also found that the initial and mean acceleration of the cut-in vehicle is less than 0.1m/s^2. For simplicity, we can reasonably assume that the cut-in vehicle maintains a constant velocity when performing lane changes. Therefore, relative velocity (v_{diff}, *m/s*), the initial relative distance ($R_{L,initial}$, *m*) and duration of lane change (T_{cut}, *s*) are selected as the decision variables describing the cut-in scenarios. The joint PDF is shown in Fig. 7.

B. Kriging Model Development and Validation

1) Car-Following Scenario: The simulation environment of the car-following scenario using the 4-dimensional decision variables is described in Fig. 8. To illustrate the validity of the Kriging method, a test matrix was generated as a benchmark.

The detailed descriptions and feasible sets for decision variables are shown in Table II. There are 81312 test cases in the Car-following Scenario-1 test matrix, and 159744 test cases in the Car-following Scenario-2 test matrix.

Each concrete scenario in the test matrix is traversed and simulated, and the enumeration results are used as benchmarks for the Kriging surrogate model verification. Finally, 1060 and 19791 collision cases were captured within the two test matrices.

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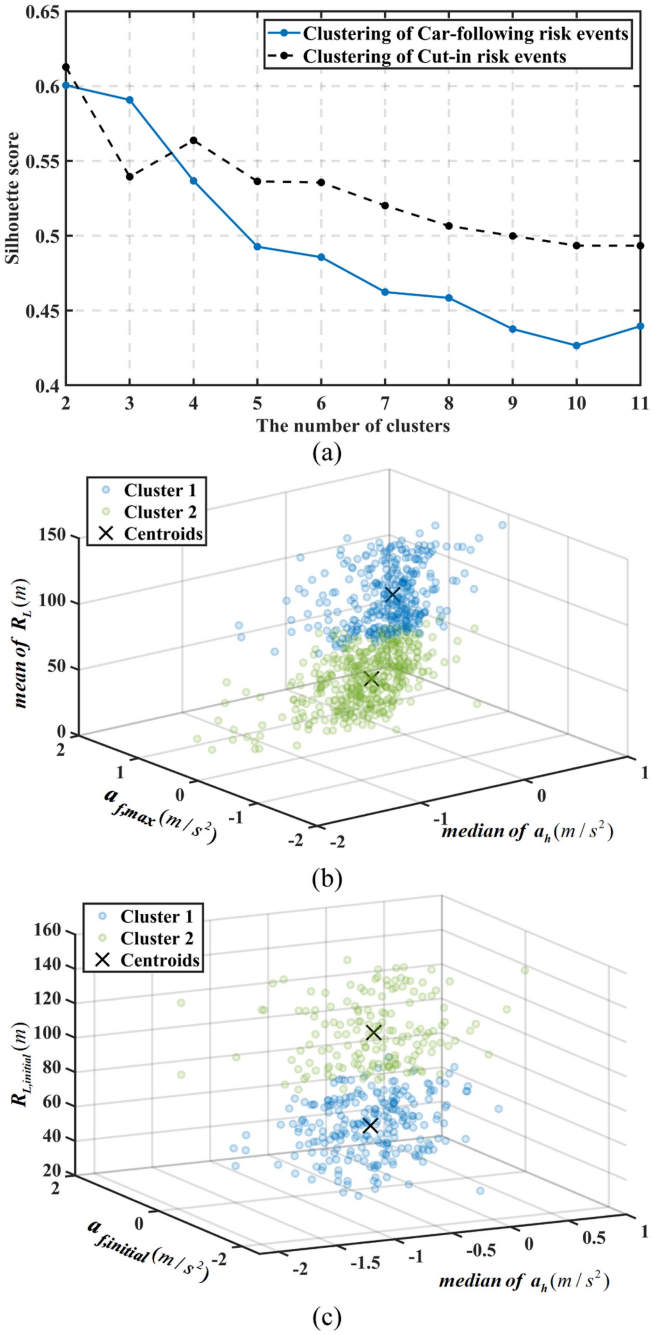


Fig. 5. Clustering results of risk car-following events and risk cut-in events. (a) Silhouette scores for various cluster numbers (b) The optimal car-following risk events clustering. (c) The optimal cut-in risk events clustering.

For the Kriging model development, 500 concrete scenarios are randomly generated from the feasible sets as the initial training set. The sequence sampling strategy stated in formulation (15) is used to iteratively select the next training sample.

Precision, recall, and F1-score were applied to evaluate the performance of the Kriging model. Take the collision rare event as an instance. The definitions of these three indicators are as follows:

- *Precision:* How many of the test cases predicted to be collisions ended up being accidents.

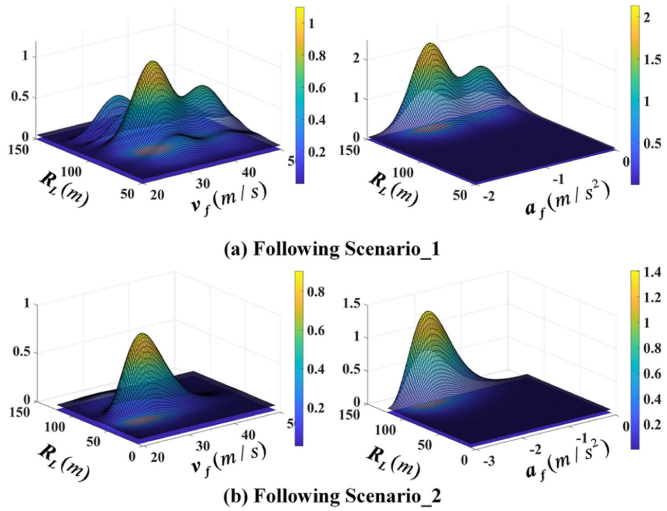


Fig. 6. The joint probability densities of the car-following decision variables (just take v_f versus R_L, and a_f versus R_L for ease of understanding).

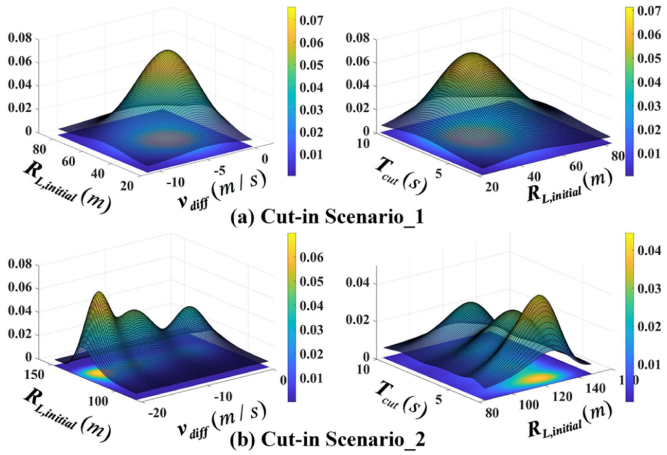


Fig. 7. The joint probability densities of the cut-in decision variables (take v_{diff} versus $R_{L,initial}$ and T_{cut} versus $R_{L,initial}$ as an example).

- *Recall:* How many collision scenarios are captured using the Kriging model.
- *F1-score:* This is a weighted comprehensive indicator of precision and recall:

$$F1-socre = 2 \times \frac{precision \times recall}{precision + recall} \tag{18}$$

In general, the larger the value of the above indicators, the better the accuracy. Table III lists the test results of the Kriging model with different training sizes.

Table III tells that the Kriging model performed well for safe cases. A model can reach an F1 score of above 0.98 with only 500 initial training cases. This is due to the fact that the safe concrete scenarios account for the majority of the training set. For collision prediction, the precision and recall increase as the training size expands. When the number of training samples reaches 1500, the Kriging model found 96.04% of the collision in car-following scenario-1, while the prediction accuracy reaches

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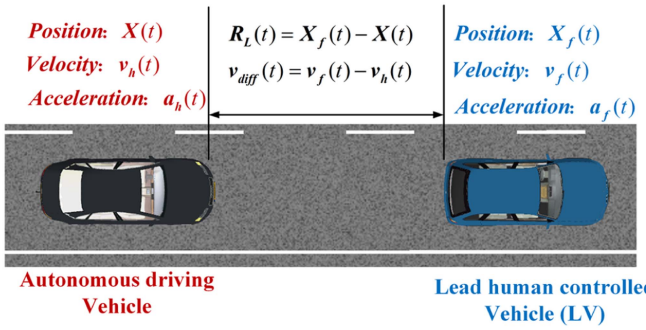


Fig. 8. Car-following simulation scenario.

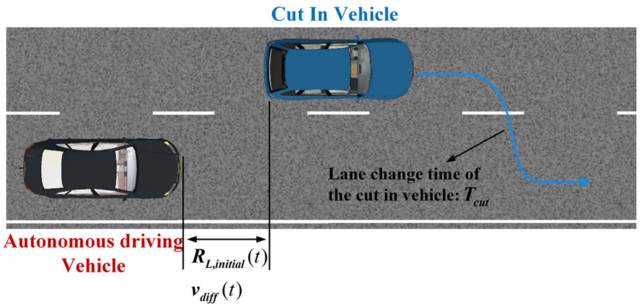


Fig. 9. Cut-in simulation scenario.

TABLE II
DECISION VARIABLES OF THE CAR-FOLLOWING SCENARIOS

Name	Description	Category	Min	Max	Offset
v_f (m/s)	Speed of LV	Following Scenario-1	20	50	2
		Following Scenario-2	20	50	
a_f (m/s^2)	Deceleration of LV	Following Scenario-1	-2	0	0.2
		Following Scenario-2	-3	0	
t_f (s)	Deceleration time of LV	Following Scenario-1	0.1	11	0.5
		Following Scenario-2	0.1	13	
R_L (m)	Initial relative distance	Following Scenario-1	50	150	5
		Following Scenario-2	7	123	

NOTE: LV is short for Leading Vehicle. Feasible sets are obtained from the statistics of driving events in the clustered car-following risk subspaces.

TABLE III
RESULTS OF KRIGING MODELS FOR CAR-FOLLOWING SCENARIOS

Category	Number of Tests	Precision	Recall	F1-score
Following Scenario-1 Safe	500	0.9920	0.9981	0.9950
	1000	0.8908	0.9986	0.9986
	1500	**0.9994**	**0.9987**	**0.9990**
Following Scenario-1 Collision	500	0.7266	0.3887	0.5065
	1000	0.8929	0.8887	0.8908
	1500	**0.9025**	**0.9604**	**0.9305**
Following Scenario-2 Safe	500	0.9812	0.9821	0.9816
	1000	0.9912	0.9855	0.9884
	1500	**0.9940**	**0.9872**	**0.9906**
Following Scenario-2 Collision	500	0.8724	0.8671	0.8697
	1000	0.9017	0.9380	0.9195
	1500	**0.9135**	**0.9579**	**0.9352**

0.9025. In car-following scenario-2, the precision and recall of collisions reach 0.9135 and 0.9579, respectively.

Compared to the enumeration benchmark, the Kriging model used only 0.94%∼1.84% (1500/159744 and 1500/81312) computational resources for training. In a word, it showed great potential for accelerating the ADS test, and the accuracy of the Kriging model is expected to be further improved with more training samples.

TABLE IV
DECISION VARIABLES OF THE CUT-IN SCENARIOS

Name	Description	Category	Min	Max	Offset
v_{diff} (m/s)	The initial relative speed between the cut-in vehicle and ego vehicle	Cut-In Scenario-1	-11	0	0.5
		Cut-In Scenario-2	-20	0	
$R_{L,\text{initial}}$ (m)	The initial relative distance between the cut-in vehicle and the ego car	Cut-In Scenario-1	22	83	2
		Cut-In Scenario-2	84	148	
T_{cut} (s)	Duration of Lane change of the cut-in vehicle	Cut-In Scenario-1	3	10	0.5
		Cut-In Scenario-2	3	10	

NOTE: There are 10,695 test cases (358 collisions) and 20,295 (3,455 collisions) test cases, respectively. Feasible sets are obtained from the statistics of driving events in the clustered cut-in risk subspaces.

TABLE V
RESULT OF KRIGING MODELS FOR CUT-IN SCENARIOS

Category	Number of Tests	Precision	Recall	F1-score
Cut-In Scenario-1 Safe	500	0.9930	0.9956	0.9943
	1000	0.9991	0.9961	0.9976
	1500	**0.9992**	**0.9991**	**0.9992**
Cut-In Scenario-1 Collision	500	0.8644	0.7988	0.8303
	1000	0.8972	0.9749	0.9344
	1500	**0.9749**	**0.9777**	**0.9763**
Cut-In Scenario-2 Safe	500	0.9723	0.9730	0.9726
	1000	0.9813	0.9765	0.9789
	1500	**0.9777**	**0.9808**	**0.9793**
Cut-In Scenario-2 Collision	500	0.8678	0.8648	0.8663
	1000	0.8883	0.9094	0.8987
	1500	**0.9091**	**0.9242**	**0.9166**

2) Cut-in Scenario: Fig. 9 describes the simulation environment of the 3-dimensional cut-in scenario.

A test matrix is also given through a cross-product of predefined decision variables, and feasible sets are summarized in Table IV.

The precision, recall, and F1 score with a different number of training samples are reported in Table V.

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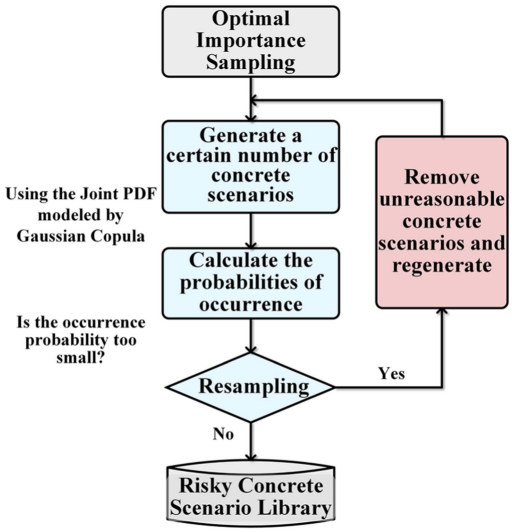


Fig. 10. The risky concrete scenario library generation process.

Table V illustrates that the Kriging model also performed well in 3-dimensional cut-in scenarios. After 1500 runs, more than 97.7% and 92% of collisions were successfully captured in cut-in scenario-1 and scenario-2, respectively. Compared to enumeration, only 7.39%∼14.03% (1500/20295 and 1500/10695) test resources were used.

C. Estimation of Car-Following Crash Rates

As described in Algorithm 1, reasonable IS parameters are obtained by solving the optimization problem shown in (16). A genetic algorithm is used to find the optimal $\boldsymbol{\mu}^*$ and $\sum^*$. For simplicity, we set the $\sum^*$ as a diagonal matrix. After 100 iterations, for car-following scenario-1, the optimal parameters $\boldsymbol{\mu}^*$ and $\sum^*$ converge to [25,-1.17,8.8131.29] and diagonal matrix (4.61,6.76,0.15,9.83); for car-following scenario-2, the optimal parameters $\boldsymbol{\mu}^*$ and $\sum^*$ converge to [31,-1.93,11.22,89.54] and diagonal matrix (4.73,0.72,5.46,8.32), respectively.

The risky concrete scenarios library is then generated using the searched reasonably IS distribution parameters. To prevent the generated test case from exceeding the feasible sets. We perform a double verification, i.e., using the obtained Gaussian Copula to calculate the occurrence probability of each test case, and those unreasonable scenarios with a quite low value are filtered out. The process of constructing a candidate risk concrete scenario library is shown in Fig. 10.

Ultimately, we only need to simulate the small-scale candidate concrete scenarios and transform the test results to the original risk subspace using the theory described in Section III.A. Fig. 11 illustrates the estimation for two types of car-following scenarios using Monte Carlo baseline and Importance Sampling techniques.

Note that the estimation results and relative errors of the Monte Carlo sampling are calculated by equations (11) and (12). The estimation relative error is shown in Fig. 12.

For car-following scenario-1, the relative error is smaller than 0.1 after running $N_{MC} \approx 8.37 \times 10^7$ and $N_{IS} \approx 12800$ times

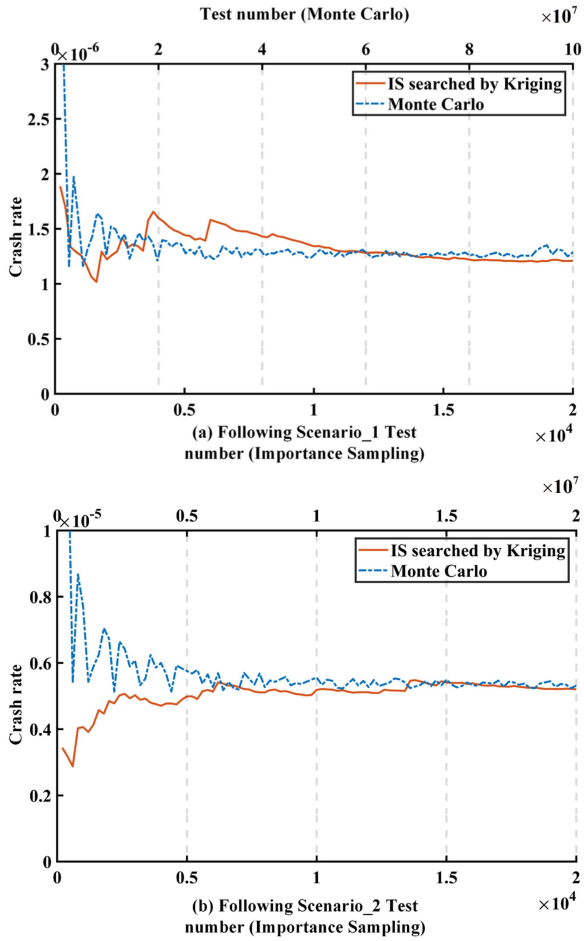


Fig. 11. Estimation of the ADS car-following crash rate. (a) In the car-following scenario-1, both the IS and the Monte Carlo crash rate are estimated to be about 1.21×10^{-6}. (b) In the car-following scenario-2, the crash rate is converged to be around 5.20×10^{-6}.

for Monte Carlo and IS, respectively. For car-following scenario-2, N_{IS} is reduced to 3800 , while Monte Carlo conducts at least 1.5×10^7 simulations. The above experiments lead to the conclusion that the Importance Sampling technique achieves a speedup of 3947 to 6539 times compared to Monte Carlo, which also shows that the Kriging surrogate model can indeed help us search for reasonable IS parameters efficiently and accurately.

D. Estimation of Cut-in Crash Rates

For cut-in scenario-1, the optimal parameters $\boldsymbol{\mu}^*$ and $\sum^*$ converge to [27, 28.98, 7.05] and diagonal matrix (1.33, 9.99, 1.77); for cut-in scenario-2, the optimal parameters$\boldsymbol{\mu}^*$ and $\sum^*$ converge to [23, 90.8, 7.93] and diagonal matrix (3.21, 9.97, 4.98), respectively. ADS crash rates within the cut-in scenarios are shown in Fig. 13.

The crash rate convergence is reached when the relative error is below 0.1. Fig. 14(a) shows that for cut-in scenario-1, the Monte Carlo and Importance Sampling achieve convergence after $N_{MC} \approx 4.8 \times 10^4$ simulations and $N_{IS} \approx 9800$ simulations, respectively. For crash rate estimation in cut-in scenario-2, Monte Carlo and Importance Sampling conduct simulations

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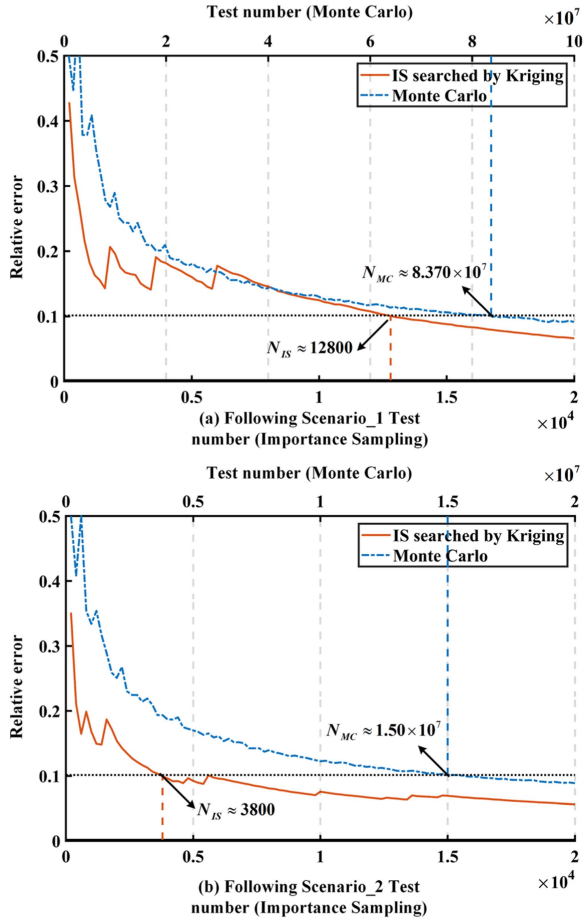


Fig. 12. Relative error of the ADS car-following crash rate. (a) The Importance Sampling distribution searched by Kriging runs only about 12800 simulations for the relative error to be less than 0.1, while Monte Carlo requires at least 8.37×10^7 test cases. (b) in the car-following scenario-2, Importance Sampling needs only 3800 test cases to give an unbiased estimation, while the Monte Carlo runs at least 1.5×10^7 simulations.

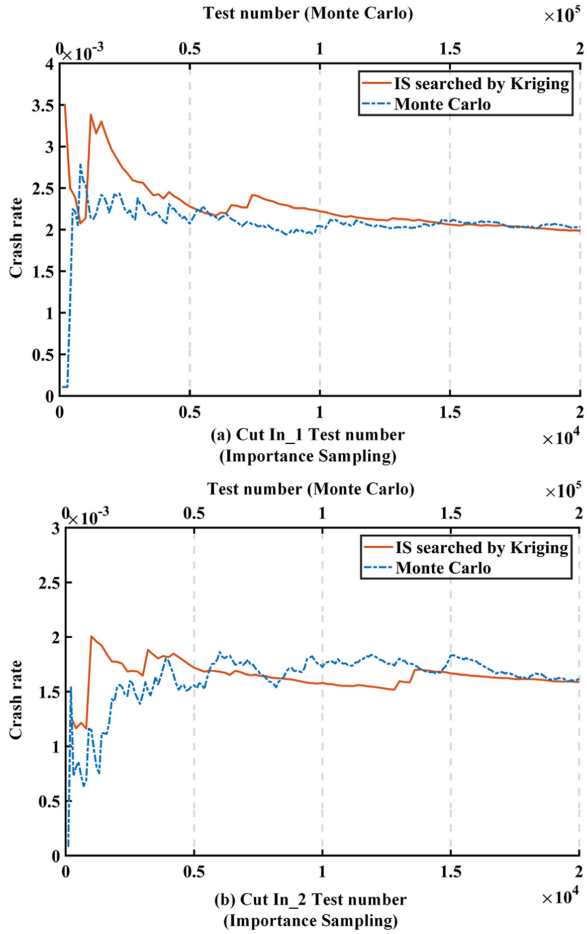


Fig. 13. Estimation of the ADS cut-in crash rate. (a) in the cut-in scenario-1, both the Importance Sampling and the Monte Carlo crash rate are estimated to be about 1.985×10^{-3}. (b) in the cut-in scenario-2, the crash rate is converged to be around 1.59×10^{-3}.

of 6.1×10^4 and 7200 to make a relative error below 0.1, as indicated in Fig. 14(b).

Specifically, the experimental results show that the safety level of ADS varies in different risk subspaces. For example, the crash rate of car-following scenario-2 is higher than that of car-following scenario-1 since the corresponding cluster-2 contains events of the leading vehicle slowing down more severely. Similarly, the ADS failure probability is higher in cut-in scenario-1 than that in cut-in scenario-2 due to the shorter initial relative distance. This reveals that obtaining more dangerous risk subspaces from NDD can help us further explore the safety boundary of ADS.

In summary, the improved efficiency of the ADS safety test benefits from the following: (1) we use data mining techniques to extract the risk subspace from the NDD, thus narrowing down the test space dramatically. (2) Importance Sampling essentially amplifies the probability of generating risky concrete scenarios, therefore a large number of low-risk test cases are skipped. (3) With the Kriging surrogate model, we can search for appropriate IS distribution parameters with little effort. Moreover, the newly introduced Gaussian Copula method allows us to more accurately model the joint PDF of the decision variables, further improving the reliability of the estimated ADS crash rates.

V. Discussion

This section reaffirms our motivation for the study, analyzing the injury severities of concrete scenarios generated by the Importance Sampling technique and other issues.

A. Why Use Naturalistic Driving Data?

To accelerate quantify the ADS probability of failure (such as crash rate). Naturalistic driving data is employed in this study rather than accident datasets for several reasons:

1) The NDD records vehicle trajectories completely and accurately. In contrast, for the accident dataset, we can only infer the kinematic characteristics through a post-accident investigation [49], which makes it difficult for us to model the joint PDF of decision variables.
2) It has been found that human factors are the major cause of crashes in the accident dataset [50]. However, ADS can eliminate driver errors (e.g., distracted driving). Therefore,

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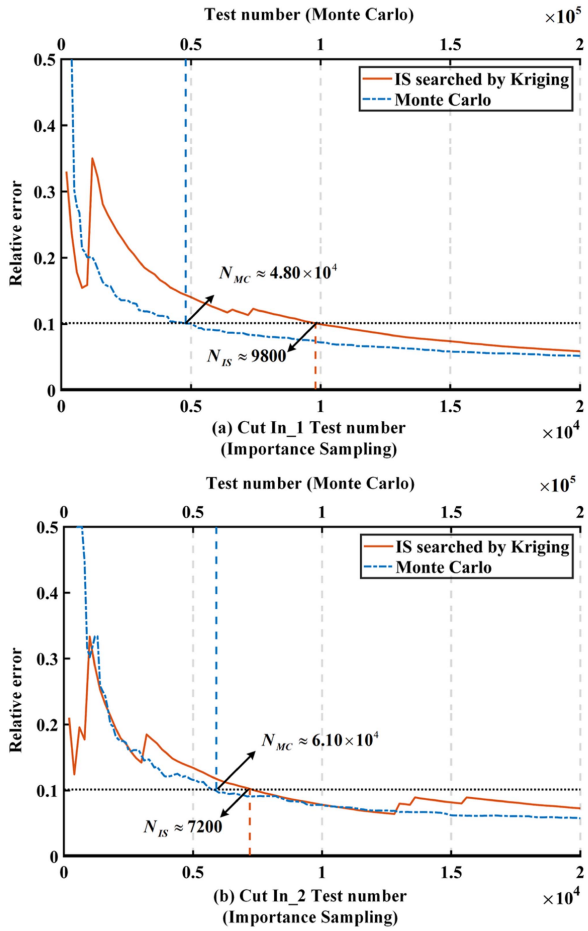


Fig. 14. Relative error of the ADS cut-in crash rate. (a) in cut-in scenario-1, the Importance Sampling runs only about 9800 simulations for the estimated relative error to be less than 0.1, while Monte Carlo requires at least 4.8×10^4 test cases. (b) for the Importance Sampling and Monte Carlo approaches, 7200 and 6.1×10^4 simulations were required in the cut-in scenario-2, respectively.

scenarios reconstructed from accident datasets may not necessarily lead to ADS failure as well.

Naturalistic driving data is, of course, limited as it will miss many of the more hazardous driving events compared to the accident database, but it does contain a large number of safety-critical driving events from which crash data can often be extrapolated [51]. In other words, testing ADS using accident datasets is more like exploring the boundaries of the ADS capabilities, whereas using naturalistic driving data yields crash rate estimation under challenging and real traffic conditions.

B. Generated Crash Scenarios Analysis

In this paper, only the relative distance between the ADS and the traffic vehicle is used as an indicator of whether a collision occurs, but a collision does not mean catastrophic consequences, such as injuries.

Krampe et al. [52] analyzed the *ISO 26262* injury severity (*S*-parameter) using the GIDAS car-to-car accident database. Specifically, the *S*-parameter gets higher with the increase in impact speed (Δv , km/h). Therefore, using the finding in [52], we count the impact speeds of various risky concrete scenario libraries and classify them into different *S*-parameters, as shown in Table VI.

TABLE VI
PERCENTAGE OF VARIOUS INJURY SEVERITY LEVELS FOR RISKY CONCRETE SCENARIOS

Category	*S*0	*S*1	*S*2	*S*3
Following Scenario-1	44.16%	38.44%	17.4%	0%
Following Scenario-2	10.21%	9.16%	26.86%	53.77%
Cut-In Scenario-1	22.31%	24.89%	49.07%	3.73%
Cut-In Scenario-2	11.69%	11.11%	29.31%	47.88%

NOTE: *S*0: cannot be defined as security-related damage ($\Delta v \leq 16$ km/h); *S*1: $16<\Delta v \leq 22$ km/h; *S*2: minor injury ($22<\Delta v \leq 33$ km/h); *S*3: serious injury ($\Delta v > 33$ km/h). These thresholds are the mean values for the belted front-row occupants in the GIDAS frontal accidents [52].

Table VI illustrates that similar to the GIDAS dataset, the proposed method can indeed generate a large number of injury concrete scenarios. It is important to note that *S*0-level concrete scenarios are not meaningless, and in fact, any form of ADS accident should be avoided at present, for it may ruin public confidence in autonomous driving technology.

C. Problems in Practical Applications

In the methodological framework of this paper, the actual executions of high-fidelity simulations focus only on the Kriging model development and the risky concrete scenario library testing. It should be noted that additional enumeration tests are just performed as a comparative benchmark to demonstrate the feasibility of the surrogate models. In practice, we terminate the Kriging model iteration by setting the training size (N_{limit}) and K-fold cross-validation. If the kriging accuracy is unstable or does not meet the requirements, the acquisition function should be improved or the N_{limit} should be increased moderately.

The final step of this paper is to generate a small number of high-risk and reasonable concrete scenarios. The actual simulation results of these test cases are obtained so that the crash rate can be estimated using the Importance Sampling theory. Moreover, the theoretical calculations can also tell us the estimation error based on the current test size (see Section III.A). Therefore, we recommend that by setting a threshold for the relative error in advance (such as 10%), we can stop the simulation if the ADS crash rate estimation error is less than the set threshold. Otherwise, we should continue to add samples to the concrete scenario library.

VI. CONCLUSION

In this paper, we present a unified and hierarchical framework of efficient and unbiased safety test for Autonomous Driving Systems (ADS) with naturalistic driving data. First, we extract the risk subspace from NDD. Second, the joint PDFs of the decision variables are subsequently modeled with the Gaussian Copula method. We then search the reasonable Importance Sampling parameters based on the Kriging model. Compared to the

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enumeration test, the Kriging surrogate model uses only limited computational resources (0.91∼1.84% for the car-following scenario and 7.39∼14.03% for the cut-in scenario) with a prediction accuracy of more than 90%. Finally, risky concrete scenario libraries are derived using the Importance Sampling parameters searched by Kriging. The results analysis show that the test efficiency is sped up to 6539 times while the estimation error does not exceed 10%.

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Zhengmin Jiang received the master's degree in automotive engineering from Chang'an University, Xi'an, China, in 2018. He is currently working toward the Ph.D. degree in pattern recognition and intelligent system with the University of Chinese Academy of Sciences, Beijing, China. From 2018 to 2020, he was a Software Engineer for SAIC Motor Technical Centre. His research interests include autonomous vehicle safety and machine learning.



Wenbo Pan received the Ph.D. degree in engineering from the University of Chinese Academy of Sciences , Beijing, China, in 2015. He is currently a Senior Engineer with CRRC Zhuzhou Institute Co, Ltd, Zhuzhou, China. His research focuses on autonomous driving environment perception.



Jia Liu received the B.E. degree in mechanical engineering from the Hefei University of Technology, Hefei, China, in 2009, the M.E. degree in vehicle engineering from the Beijing Institute of Technology, Beijing, China, in 2015, and the Ph.D. degree in pattern recognition and intelligent systems from the University of Chinese Academy of Sciences, Beijing, China, in 2020. He is currently an Associate Professor with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Beijing, China. His research interests include intelligent control and reinforcement learning.



Shaobo Dang received the bachelor's and master's degrees in engineering from Northwestern Polytechnical University, Xi'an, China, in 2011 and 2014, respectively, and the Ph.D. degree in computer science from the University of New South Wales, Sydney, NSW, Australia, in 2018, He is currently an Assistant Research Scientist with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Beijing, China. His research focuses on the Internet of Vehicles.



Zhiheng Yang received the master's degree in control engineering from the Guilin University of Electronic Science and Technology, Guilin, China, in 2018. He is currently an Engineer with the Research Center for Automotive Electronics, Shenzhen Institute of Advanced Technology, Shenzhen, China. His research interests include computer vision and deep learning.



Huiyun Li (Senior Member, IEEE) received the M.Eng. degree in electronic engineering from Nanyang Technological University, Singapore, in 2001, and the Ph.D. degree from the University of Cambridge, Cambridge, U.K., in 2006. She is currently a Professor with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Beijing, China. Her research focuses on automobile electronics.



Yi Pan (Senior Member, IEEE) received the B.E. and M.E. degrees in computer engineering from Tsinghua University, Beijing, China, in 1982 and 1984, respectively, and the Ph.D. degree in computer science from the University of Pittsburgh, Pittsburgh, PA, USA, in 1991. He is currently a Chair Professor and the Dean of the College of computer science and control engineering, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Beijing, and a Regents' Professor Emeritus with Georgia State University, Atlanta, GA, USA. His work has been cited more than 20000 times in Google Scholar and H-index is 90. His current research interests include big data analytics, cloud computing, and machine learning technologies.

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